

SDI Training for ICT Professionals

# KoboToolbox Overview

Mobile Data Collection for Fieldwork

**Masoud Hamad (SUZA)** | **Msilikale Msilanga (UTU)**

Day 3 | 1 Hour Lecture | [crd.resilienceacademy.ac.tz](http://crd.resilienceacademy.ac.tz)

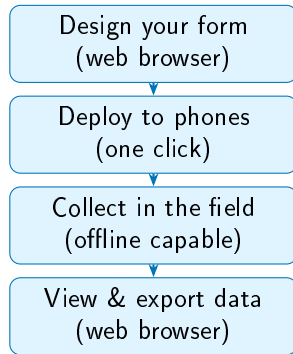
# What is KoboToolbox?

## A Free Tool for Collecting Data in the Field

- **Open-source** platform for mobile data collection
- Used by **600,000+** organisations worldwide
- Created at Harvard for humanitarian work in challenging environments
- Supported by **OCHA** (UN Office for Coordination of Humanitarian Affairs)

## Key Advantage

Collect data on your phone **even without internet** – GPS coordinates, photos, survey answers – and



## Real-World Example: Tanga Flood Mapping

### Resilience Academy / GIZ Project

**16 Resilience Academy alumni** used KoboToolbox to collect flood data in Tanga, Tanzania.

#### What they collected:

- GPS coordinates of flooded areas
- Photos of flood damage
- Water depth estimates
- Infrastructure impact (roads, buildings)
- Timestamps for each observation

#### How they did it:

- Installed **KoboCollect** app on Android phones
- Walked through flood-affected neighborhoods
- Filled in survey forms at each location
- Phone recorded GPS automatically
- Uploaded data when back at the office

### Result

## Use Cases: Where KoboToolbox Fits in Government Work

### Disaster Assessment

- Damage surveys after floods, storms
- Record locations of affected buildings
- Photo evidence for insurance / aid
- Real-time situation reports

### Health Surveys

- Household health assessments
- Water quality sampling locations
- Vaccination coverage tracking
- Facility condition surveys

### Infrastructure Audits

- Map condition of roads and bridges
- GPS-locate water points and wells
- Record status of public buildings
- Track maintenance needs

### Environmental Monitoring

- Forest cover observation points
- Coastal erosion measurements
- Waste dump site mapping
- Mangrove restoration tracking

## Cloud vs Self-Hosted: Why Run Your Own Server?

### Option A: Use [kobo.humanitarianresponse.info](https://kobo.humanitarianresponse.info)

- Free cloud service by OCHA
- No installation needed
- Sign up and start immediately
- × Data stored outside Tanzania
- × No control over server
- × Subject to usage limits
- × Depends on external availability

### Option B: Self-Hosted (What We Will Do)

- ✓ **Data sovereignty** – data stays in your country
- ✓ **Government compliance** – meets local data policies
- ✓ **Full control** – you manage users and backups
- ✓ **No usage limits** – unlimited submissions
- ✓ **Custom branding** – your organisation's name
  - Requires a server with Docker

# KoboCollect App: Offline GPS Data Collection

## What is KoboCollect?

A free Android app that lets field workers fill in survey forms on their phones – **no internet needed** during data collection.

## How it works:

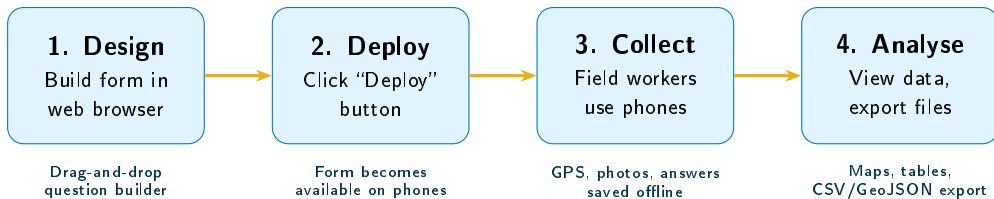
- 1 Open **KoboCollect** on your Android phone
- 2 Tap “**Get Blank Form**” (needs internet once)
- 3 Go to the field – **no internet needed**
- 4 Fill in forms, phone captures GPS automatically
- 5 Take photos directly within the form
- 6 When back online, tap “**Send Finalized Form**”

## KoboCollect

- Fill Blank Form
- Edit Saved Form
- Send Finalized Form
- Get Blank Form
- Delete Saved Form

Works offline  
GPS auto-capture

# The Complete Workflow: Form → Deploy → Collect → Analyse



## What You Need

- A computer with a web browser (for steps 1, 2, 4)
- Android phones with KoboCollect (for step 3)
- Internet access at office (not needed in the field)

## No Programming Required

- Form design is visual (drag and drop)
- Deployment is one click
- Data export is done through menus
- Everything happens in a web browser

## Live Demo: What You Will See

The instructor will now demonstrate KoboToolbox live on the projector.

Follow along on screen – you will do this yourself in the practical session.

### Demo steps:

- 1 **Login page** – Open the KoboToolbox web interface in a browser
- 2 **Dashboard** – See the list of forms (projects)
- 3 **Open a deployed form** – The Tanga Flood Survey form
- 4 **View submissions** – See the data table with all collected entries
- 5 **Map view** – See GPS points plotted on a map
- 6 **Single submission** – Click one entry to see its details and photo
- 7 **Export menu** – Show the download options (CSV, Excel, GeoJSON)

### What to Notice

- Every submission has a **GPS coordinate** – that is what makes this data *spatial*

## Key Takeaways and What Comes Next

### What We Learned

- KoboToolbox is a **free, open-source** tool for collecting data in the field
- It captures **GPS coordinates** automatically – making your data spatial
- The **KoboCollect app** works offline on Android phones
- Self-hosting keeps **data under your control**
- The workflow is simple: **Design → Deploy → Collect → Analyse**

### Coming Up Today

- S2 Installation** – Set up KoboToolbox on your training server using Docker
- S3 Form Design** – Build the Tanga Flood Survey form step by step
- S4 Administration** – Manage users, export data, run backups
- S5 Integration** – Move your Kobo data into QGIS and the CRD